

MEDIRAG: A Retrieval-Augmented Generation Based Medical Question Answering Assistant

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Abstract—In modern digital healthcare, open-web search engines and unconstrained large language models (LLMs) frequently deliver unverified, incomplete, or hallucinated medical advice. Such inaccuracies pose severe risks to patient health and clinical decision-making. To mitigate these hazards, this paper presents **MEDIRAG**, an end-to-end web-based Medical Question Answering (QA) Assistant grounded in trusted biomedical knowledge repositories. MEDIRAG integrates the National Institutes of Health (NIH) MedQuAD dataset as its primary knowledge source, containing 14,978 curated medical question-answer pairs. The system utilizes BioBERT embeddings (768-dimensional) to capture deep biomedical semantics and indexes them within a High-Performance FAISS Vector Database using cosine similarity (*IndexFlatIP*). A hybrid retrieval pipeline—combining semantic search, lexical keyword matching, medical topic classification, and intent detection—ranks candidate sources. The retrieved context is processed using LangChain and a locally hosted Qwen3 (1.7B) LLM via Ollama to generate precise, grounded answers accompanied by verifiable source attribution URLs. Equipped with a React/Vite web interface, Flask REST API, user authentication, query history tracking, and safety rejection guardrails, MEDIRAG bridges the gap between complex biomedical AI architectures and accessible, trustworthy healthcare platforms.

Index Terms—Retrieval-Augmented Generation (RAG), BioBERT, FAISS, MedQuAD, LangChain, Qwen3, Biomedical Question Answering, Clinical AI, Fact Grounding.

1 INTRODUCTION

1.1 Background

The proliferation of digital health portals has transformed how patients and medical practitioners seek information regarding symptoms, diagnoses, and treatment options. While web accessibility has improved health literacy, it has also amplified the dissemination of unverified medical information. General-purpose Natural Language Processing (NLP) models and Large Language Models (LLMs) demonstrate fluent conversational capabilities; however, they inherently suffer from “hallucination”—generating factually incorrect or unsupported claims with high confidence. In biomedical contexts, hallucination can lead to misdiagnoses, inappropriate self-medication, and delayed clinical intervention.

1.2 Problem Statement

Standard generative AI architectures lack parametric memory boundaries and continuous updates from validated clinical sources. Many existing healthcare QA tools either generate responses purely from closed-weight LLM parametric knowledge without explicit source citations, or rely solely on traditional keyword-matching search engines that fail to comprehend complex biomedical queries. There is an urgent need for an AI-assisted question-answering architecture that dynamically retrieves information from authoritative medical repositories prior to response synthesis, ensuring answers are grounded, transparent, and verifiable.

1.3 Research Gap

Existing medical QA literature reveals critical limitations:

- 1) **Lack of Retrieval Grounding:** Standalone LLMs (e.g., standard BioBERT fine-tunes) generate answers without dynamic document retrieval or citation mechanisms.
- 2) **High Computational Barriers & Proprietary Closures:** State-of-the-art models like Med-PaLM achieve strong accuracy but remain proprietary and computationally prohibitive for open deployment.
- 3) **Absence of Full-Stack Usability:** Research-focused RAG models (e.g., i-MedRAG) often exist strictly as offline benchmark scripts, lacking user authentication, interactive dashboards, source traceability, and real-time safety fallback mechanisms.

1.4 Research Objectives

- **RO1:** Build a full-stack, web-based Medical Question Answering platform using Flask REST API and React/Vite/TypeScript frontend.
- **RO2:** Curate and index 14,978 clean medical QA records from the MedQuAD dataset.
- **RO3:** Implement a 768-dimensional semantic embedding space using BioBERT combined with a FAISS vector index (*IndexFlatIP*).
- **RO4:** Develop a hybrid retrieval engine integrating semantic similarity, keyword matching, topic extraction, and intent classification.
- **RO5:** Construct a constrained RAG pipeline with LangChain, Ollama, and Qwen3 (1.7B) to enforce grounded answer generation and reject queries lacking sufficient evidence.

1.5 Research Questions

- **RQ1:** How significantly does BioBERT-FAISS semantic search improve medical context retrieval accuracy compared to traditional keyword matching?
- **RQ2:** Can a local, parameter-efficient LLM (Qwen3 1.7B) constrained by RAG eliminate parametric hallucinations in biomedical QA?
- **RQ3:** Does a multi-stage hybrid retrieval mechanism effectively resolve ambiguous medical queries?

1.6 Contributions

- **Grounded Biomedical Pipeline:** An open-source RAG architecture that maps queries directly to NIH-verified MedQuAD sources.
- **Hybrid Retrieval Mechanics:** Integration of BioBERT dense vectors and FAISS similarity search with explicit query intent detection and topic matching.
- **User Safety & Attribution:** Built-in citation tracking (displaying source URLs and relevance scores) alongside an automated safety check that rejects out-of-domain or ungrounded queries.
- **Full-Stack Implementation:** A complete responsive Web application with glassmorphic UI, search history persistence via SQLite, and user authentication.

1.7 Organization of the Paper

The rest of this paper is organized as follows: Section 2 provides a comprehensive literature survey and comparative analysis. Section 3 details the MEDIRAG methodology, data preprocessing, mathematical foundations, and system design. Section 4 describes the experimental setup and evaluation metrics. Section 5 presents empirical results and UI executions. Section 6 discusses clinical implications, while Sections 7 and 8 outline limitations, conclusions, and future research directions.

2 RELATED WORK

2.1 Existing Approaches

Biomedical NLP has transitioned from dictionary-based information retrieval to semantic vector-space representations. Early systems relied on static biomedical knowledge graphs (e.g., UMLS) and rule-based keyword matchers, which struggled with paraphrased user queries.

2.2 Machine Learning & Deep Learning Approaches

The introduction of Transformer architectures led to domain-specific pre-trained models such as BioBERT. Lee et al. (2019) demonstrated that BioBERT, pre-trained on PubMed abstracts and PMC full-text articles, drastically improved named entity recognition and relation extraction. However, fine-tuned BioBERT models alone are limited to task-specific classification or extractive QA and cannot synthesize cohesive natural language answers from external documents without retrieval coupling.

2.3 LLMs & Explainability Techniques

Singhal et al. (2023) developed Med-PaLM, demonstrating physician-level performance on medical licensing examinations. Despite its accuracy, Med-PaLM relies on massive proprietary parametric weights without real-time external source linkage. Recent efforts like i-MedRAG (2024) and FAISS-BioBERT RAG implementations (2025) demonstrate that coupling LLMs with dynamic retrievers significantly boosts domain accuracy while reducing model footprint.

2.4 Critical Comparison

Table 1 summarizes existing literature against the proposed MEDIRAG framework.

3 PROPOSED METHODOLOGY

3.1 Overall Framework

MEDIRAG follows a five-layer modular architecture:

- 1) **User Interface Layer (Frontend):** React + Vite + TypeScript application providing interactive query input, glassmorphic UI, search history, and citations.
- 2) **Application Layer (Backend):** Python Flask REST API managing authentication, request routing, and payload validation.
- 3) **Retrieval Layer:** BioBERT encoder (768 dimensions) paired with a FAISS vector index and hybrid keyword/intent matchers.
- 4) **Generation Layer:** LangChain prompt orchestrator paired with Ollama running Qwen3 (1.7B) locally.
- 5) **Output Layer:** Structured response with answer summary, evidence breakdown, URL source, and relevance scoring.

3.2 Dataset Processing

The system utilizes the **MedQuAD** dataset compiled from 12 trusted NIH institutes (e.g., NHLBI, NIDDK, GARD, Cancer.gov).

- **Raw Size:** Multi-XML structure containing question-answer pairs, categories, and source links.
- **Preprocessing Pipeline:** Stripped XML tags, removed duplicate or incomplete records, normalized biomedical terminology, and extracted official reference URLs.
- **Final Knowledge Base:** 14,978 standardized biomedical QA records.

3.3 Embeddings & Indexing Construction

Medical texts are converted into dense vector representations using BioBERT (dmis-lab/biobert-v1.1). For a given question-answer pair d_i , BioBERT maps the text into a 768-dimensional embedding space:

$$\vec{v}_i = \text{BioBERT}(d_i) \in \mathbb{R}^{768} \quad (1)$$

The vectors are normalized to unit length and indexed using FAISS `IndexFlatIP` (Inner Product, equivalent to cosine similarity for normalized vectors):

$$S_{\cosine}(q, d_i) = \frac{\vec{q} \cdot \vec{d}_i}{\|\vec{q}\| \|\vec{d}_i\|} \quad (2)$$

TABLE 1
Critical Comparison of Medical Question Answering Systems

Study / Model	Key Techniques	Primary Purpose	Identified Limitations	MEDIRAG Advantage
BioBERT Original (2019)	BioBERT, NLP	Biomedical text understanding	No document retrieval or RAG workflow	Integrates BioBERT within active FAISS retriever
BioASQ (2019)	QA, BioBERT, Fine-Tuning	Biomedical QA	Lacks semantic retrieval & source citations	Provides explicit URL attribution & dynamic context
Med-PaLM (2023)	Proprietary LLM	High-accuracy medical generation	Proprietary, high compute requirement	Fully local, open-source execution via Qwen3 + Ollama
i-MedRAG (2024)	RAG, LLM	Retrieval-augmented medical QA	Complex architecture, slow processing times	Optimized local hybrid FAISS index for rapid response
Efficient (2025)	RAG, BioBERT, FAISS	Information retrieval QA	Research focused; lacks complete web application	Complete production web application with modern UI

3.4 Hybrid Retrieval Algorithm

To maximize recall and precision, MEDIRAG combines vector search with lexical filters:

- 1) **Semantic Search:** FAISS retrieves top- K candidate vectors based on cosine distance.
- 2) **Lexical Keyword Extraction:** Regular expressions strip stop words to isolate core medical terms.
- 3) **Intent Detection:** Categorizes user intent into *Symptoms, Causes, Diagnosis, Treatments, or Prevention*.
- 4) **Hybrid Score Calculation:**

$$\text{Score}_{\text{final}} = \alpha \cdot S_{\text{semantic}} + \beta \cdot S_{\text{keyword}} + \gamma \cdot S_{\text{intent}} \quad (3)$$

Where $\alpha = 0.6$, $\beta = 0.2$, and $\gamma = 0.2$.

3.5 Grounded RAG Prompting & Generation

The retrieved context is injected into a strict system prompt executed by Qwen3 (1.7B) via LangChain and Ollama:

```
``Use ONLY the retrieved medical context below to answer the user's question. Do not assume or generate information outside this context. If the retrieved context is insufficient, state clearly that reliable information is unavailable.``
```

4 EXPERIMENTAL SETUP

4.1 Hardware and Software Environment

- **Operating System:** Windows 11 Home / Linux Ubuntu 22.04 LTS
- **CPU/GPU Environment:** Intel Core i7 / AMD Ryzen 7, 16GB RAM
- **Language Runtime:** Python 3.10+, Node.js v18+
- **Core Libraries:** Flask REST API, LangChain, FAISS-cpu, PyTorch, Transformers (dmis-lab/biobert-v1.1)
- **LLM Host Engine:** Ollama running Qwen3 (1.7B parameter model)
- **Frontend Stack:** React, Vite, TypeScript, Tailwind CSS

4.2 Database Configuration

- **Vector Store:** FAISS IndexFlatIP containing 14,978 vectors of dimension size 768.
- **Relational Database:** SQLite for managing user login accounts, password hashes, and persistent inquiry logs.

5 RESULTS

5.1 System Execution & Workflow

The system was verified across various clinical inquiry scenarios.

5.1.1 Test Query 1: "What are the symptoms of asthma?"

- **Retrieved Primary Source:** NIH MedQuAD (NHLBI)
- **Vector Match Score:** 0.88 (88% Semantic Match)
- **Topic Score:** 1.0 | **Intent Score:** 1.0 (Type: Symptoms)
- **Generation Response Time:** 75.64 seconds (local CPU compilation)
- **Output:** "The symptoms of asthma include coughing, wheezing, chest tightness, and shortness of breath..." paired with direct source URL links to Medline-Plus/NHLBI.

5.1.2 Test Query 2: Out-of-Domain Guardrail

- **Query:** "tell me about eye allergy" / "what is eye dry"
- **System Action:** System computed retrieval confidence below predefined safety thresholds (Score < 0.50).
- **Output:** Safely rejected ungrounded generation with warning: "Sufficient information was not found in the retrieved medical sources. This information is for educational purposes only."

6 DISCUSSION

6.1 Clinical & Technical Implications

- 1) **Elimination of Hallucinations:** By restricting Qwen3 generation strictly to retrieved MedQuAD text chunks, MEDIRAG eliminates parametric hallucinations.
- 2) **Traceability:** Every response displays the originating NIH body and reference URL, allowing users to verify claims independently.

- 3) **Privacy & Edge Security:** Hosting Qwen3 locally via Ollama ensures patient health queries are not transmitted to external commercial cloud servers.

7 LIMITATIONS

- 1) **Hardware Bottlenecks:** Local LLM generation on CPU architectures requires ~60–75 seconds per query; hardware acceleration (NVIDIA CUDA GPU) is required for real-time throughput.
- 2) **Knowledge Base Boundaries:** The system's domain is limited to the 14,978 records present within MedQuAD. Unindexed novel medical topics trigger automated safety rejections.

8 CONCLUSION AND FUTURE WORK

8.1 Conclusion

MEDIRAG successfully demonstrates a Retrieval-Augmented Generation platform tailored for medical question answering. By pairing BioBERT semantic representations and FAISS vector indexing with a locally deployed Qwen3 1.7B LLM, the platform delivers accurate, context-grounded, and citation-backed health information.

8.2 Future Work

- **GPU Acceleration:** Deploying tensor optimization libraries (vLLM or TensorRT-LLM) to reduce response latency below 2 seconds.
- **Knowledge Expansion:** Integrating PubMed Central and clinical guidelines (e.g., WHO, CDC) to expand index coverage beyond MedQuAD.
- **Multi-Modal QA:** Extending the architecture to process clinical imaging (X-rays, CT scans) alongside text queries.

DECLARATIONS

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Conflict of Interest: The authors declare no conflicts of interest.

Ethics Statement: This project uses publicly available, anonymized datasets (MedQuAD) and does not involve human subjects or private health records.

Data Availability: MedQuAD dataset is publicly accessible via NIH/GitHub repositories.

Author Contributions: K. Navya designed the RAG architecture and backend Flask integration; N. Deepika handled vector indexing, BioBERT embeddings, and database setup; K. Bhavya developed the React frontend UI.

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